

3.2.2 Electromagnetic radiation and quantum phenomena

3.2.2.1 The photoelectric effect

Content	Opportunities for skills development
Threshold frequency; photon explanation of threshold frequency. Work function ϕ, stopping potential. Photoelectric equation: $hf = \phi + E_{k(\max)}$ $E_{k(\max)}$ is the maximum kinetic energy of the photoelectrons. The experimental determination of stopping potential is not required.	PS 3.2 / MS 2.3 Demonstration of the photoelectric effect using a photocell or an electroscope with a zinc plate attachment and UV lamp.

3.2.2.2 Collisions of electrons with atoms

Content	Opportunities for skills development
Ionisation and excitation; understanding of ionisation and excitation in the fluorescent tube. The electron volt. Students will be expected to be able to convert eV into J and vice versa.	

3.2.2.3 Energy levels and photon emission

Content	Opportunities for skills development
Line spectra (eg of atomic hydrogen) as evidence for transitions between discrete energy levels in atoms. $hf = E_1 - E_2$ In questions, energy levels may be quoted in J or eV.	AT j / MS 0.1, 0.2 Observation of line spectra using a diffraction grating.

3.2.2.4 Wave-particle duality

Content	Opportunities for skills development
Students should know that electron diffraction suggests that particles possess wave properties and the photoelectric effect suggests that electromagnetic waves have a particulate nature. Details of particular methods of particle diffraction are not expected. de Broglie wavelength $\lambda = \frac{h}{mv}$ where mv is the momentum. Students should be able to explain how and why the amount of diffraction changes when the momentum of the particle is changed. Appreciation of how knowledge and understanding of the nature of matter changes over time. Appreciation that such changes need to be evaluated through peer review and validated by the scientific community.	PS 1.2 Demonstration using an electron diffraction tube. MS 1.1, 2.3 Use prefixes when expressing wavelength values.